

CHAPTER ONE

INTRODUCTION

The geochemical composition of clastic sedimentary rocks is intricately influenced by a multitude of factors. These factors encompass the composition of the parent rock, the processes of weathering, the climatic conditions prevailing during the weathering phase, the mechanisms of transportation, the specific depositional environment of the basin, and the subsequent post-depositional processes (e.g., diagenesis, metamorphism) (Hayashi et al., 1997).

As a result, the geochemistry of clastic sediments serves as a crucial indicator of their provenance, the extent of chemical weathering they have undergone, and the degree of hydraulic sorting they have experienced. In-depth investigations conducted across various scientific studies have provided substantial evidence that substantiates and elucidates the aforementioned aspects pertaining to the formation and genesis of both ancient and modern siliciclastic sediments (e.g., Dickenson et al., 1983; Nesbitt and Young, 1982, 1984; Bhatia, 1983; Roser and Korsch, 1988; McCann, 1991; Condie et al., 1992; Condie, 1993; McLennan et al., 1993; Nesbitt et al., 1996; Cullers, 2000).

The geochemical composition of terrigenous sedimentary rocks is influenced by several interconnected factors, including provenance, weathering, transportation, and diagenesis (Bhatia, 1983). These variables play a crucial role in shaping the chemical makeup of these rocks. Major element discrimination diagrams, such as those developed by Bhatia (1983); Roser and Korsch (1986, 1988), have been widely utilized to differentiate the tectonic settings of sedimentary basins.

There has been a notable dearth of research focused on the examination of exposed sediments in Cox's Bazar District, particularly concerning their geochemical properties. This lack of attention and investigation has resulted in a significant gap in our understanding of the geochemical composition and characteristics of sediments in this region. Further studies addressing this research gap are necessary to gain a more comprehensive understanding of the geochemistry of

sediments in Cox's Bazar District and its implications for broader geological and environmental processes.

The aim of this study is to analyze the major element geochemistry of the exposed sediments at Cox's Bazar District to interpret provenance, source area weathering, and tectonic setting of the basin. Different chemical parameters and discriminant function diagrams have been used to interpret these aims.

1.1 Background Study

Cox's Bazar District, located in southeastern Bangladesh, encompasses a geologically diverse region that offers an excellent opportunity to study the major element geochemistry of exposed sedimentary rocks. The district is characterized by a complex tectonic history, including the presence of the Chittagong Fold Belt, which has undergone significant deformation and uplift. The geological formations in Cox's Bazar District consist of a variety of sedimentary rocks, such as sandstones, shales, limestones and conglomerates, which hold valuable information about Earth's past.

The earlier works (e.g. Rahman and Suzuki, 2007, Hossain et al., 2010) focused on geochemical provenance of sub-surface Neogene Surma Group encountered from different exploration wells in the Bengal basin. Recent investigations on geochemical characteristics of ancient and modern detritus have been carried out in order to infer the source rocks, provenance and tectonic setting (e.g. Bhatia, 1983; Bhatia and Crook, 1986; Roser and Korsch, 1986, 1988; McCann, 1991; McLennan et al., 1993; Nesbitt and Young, 1996; Hayashi et al., 1997). Several studies have also been focused on the identification of palaeotectonic settings of provenances based on geochemical signatures of siliciclastic rocks (e.g., Dickinson and Suczek, 1979; Bhatia, 1983; Bhatia and Crook, 1986; Roser and Korsch 1986; McLennan and Taylor, 1991). Among the terrigenous sedimentary rocks, shales are considered to represent the average crustal composition of the provenance much better than any other siliclastic rocks (e.g., McCulloch and Wasserburg, 1978). Shales retain most of the mineral constituents of the source and their bulk chemistry preserves the near. original signature of the provenance and more faithfully reveal palaeoweathering conditions (e.g., Pettijohn, 1975). The geochemical characteristics proved the

existence of the active continental margin setting for the Neogene mud rocks of Sitakund anticline, Bengal Basin and different concentrations and ratios of major oxides satisfy that the mud rocks were derived from felsic source rocks (Gazi, et al., 2017). The geochemical characteristics suggest an active continental margin to passive margin setting for the Surma Group sandstones preserve the signatures of a recycled provenance that is agreement with sandstone petrography and derivation of these sandstones from felsic source rocks. (Rahman, et al., 2014). Geochemically, the sandstones are classified as sub-litharenite that grades to quartzarenite. Surma Group sandstones are rich in quartz followed by lithic grains, feldspar and mica (predominantly white mica). The sandstones are dominantly quartzolitic and quartzose one in composition with abundant low-grade metamorphic, sedimentary lithics, low feldspars and little volcanic detritus, indicating that the sands were derived from a quartzose recycled orogenic province. Geochemically, the sandstones are classified mainly as litharenites. The geochemical characteristics suggest an active continental margin to passive margin setting for the Surma Group sandstones, and preserve the signatures of a recycled provenance. (Rahman, et al., 2007). Geochemical classification of the sediments shows mostly subarkose with few sublitharenites. Some major and trace elements display comprehensible correlation with Al_2O_3 confirming their possible hydraulic fractionation. The chemical index of alteration (CIA*), CIW* index, index of compositional variability (ICV), values and the ratio of SiO_2/AlO_3 , suggest low degrees of chemical weathering in the source areas as well as immature to moderately mature the sediments. Whole rock geochemistry and discrimination diagrams demonstrate the continental signature derivatives, which might have been derived from the felsic to intermediate igneous rocks (granitic plutonic rocks) as well as from quartzose sedimentary/metamorphic provenance. These typical sources are present in a vast region of the Himalayan belt and catchment areas of the Ganges. The tectonic setting of the sediments demarcates typically passive margin with a slightly continental arc system (Hossain, et al., 2014). The present research examines the geochemistry of exposed muddy sediments of Cox's Bazar constrain their source rocks, tectonic setting as well as provenance and source area weathering of the studied sample.

1.2 Aims and Objectives

The main purpose of this study is to evaluate the major element geochemistry of the exposed sedimentary rocks of Cox's Bazar district, southeastern Bengal basin to their provenance and source area weathering based on geochemistry.

The objectives of the study are as follows:

- Attempts to analyze the provenance.
- Source area weathering and hydraulic sorting processes.
- To Interpret Tectonic Setting.
- To interpret Sediments maturity.

1.3 Location

Cox's Bazar District is located in the southeastern part of Bangladesh, specifically in the last southeast portion of the country. The geographical coordinates of the study area lie between N 20° 51' 9.3" to N 21° 43' 50.62" latitudes and E 91° 50' 45.36" to E 92° 18' 22.32" longitudes. The district is bounded by the world's longest sandy beach, stretching approximately 120 km along the coastline. It is characterized by the presence of Moheshkhali, Inani, and DakshinNila anticlines, with the study area encompassing Maheshkhali, Cox's Bazar Sadar, Ramu, Ukhia, Teknaf, and part of Chakariaupazila. Within the study area, the Inani structure consists of low hillocks that trend NNW-SSE and reach a maximum elevation of 542 feet above sea level, with the northern plunging area bisected by Reju Khal.

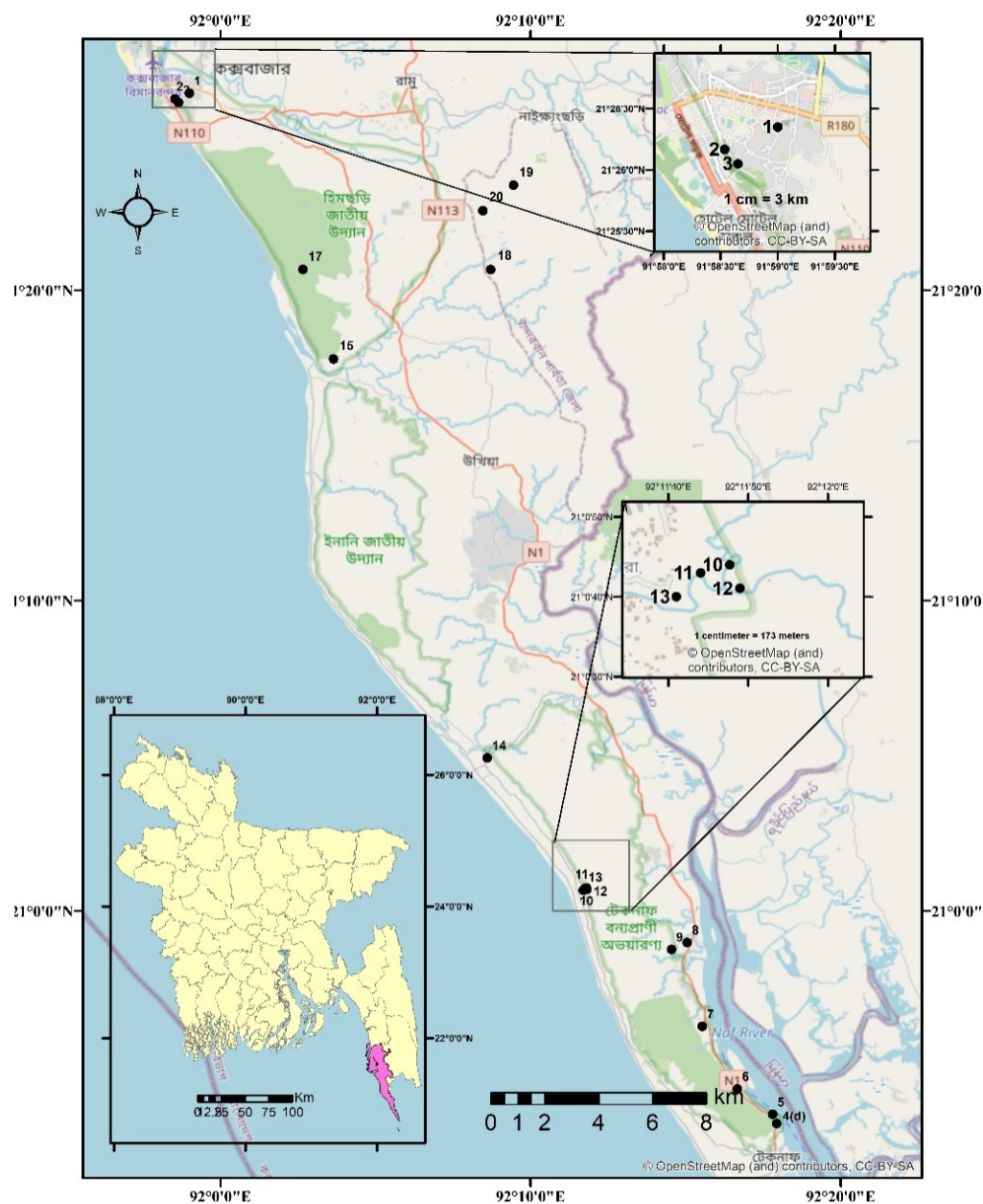


Figure 1.1: Location map of the study area

1.4 Physiography

The physiography of Cox's Bazar District in Bangladesh is characterized by diverse landforms and coastal features. The district is situated in the southeastern part of the country, along the Bay of Bengal. The most prominent feature is the long sandy beach, which stretches for about 120 km along the coastline. This sandy beach is considered the world's longest and attracts a significant number of tourists. The beach is a major part of the physiographic profile and is known for its scenic beauty. In addition to the sandy beach, the district features various other landforms. These include low-lying plains, hills, and hillocks. The Moheshkhali, Inan, and Dakhin Nila anticlines are notable geological structures found in the area. The Inani structure, characterized by NNW-SSE trending low hillocks, divides the northern plunging area with the presence of RejuKhal. These features contribute to the overall physiographic diversity of the district. The district's physiography is influenced by its location along the Bay of Bengal, experiencing the effects of coastal erosion and deposition. It is important to note that the physiography of Cox's Bazar District is subject to natural processes and can be influenced by human activities.

Overall, the physiography of Cox's Bazar District presents a unique blend of coastal features, including the extensive sandy beach, along with hills, plains, and anticlines, contributing to the diverse landscape of the region.

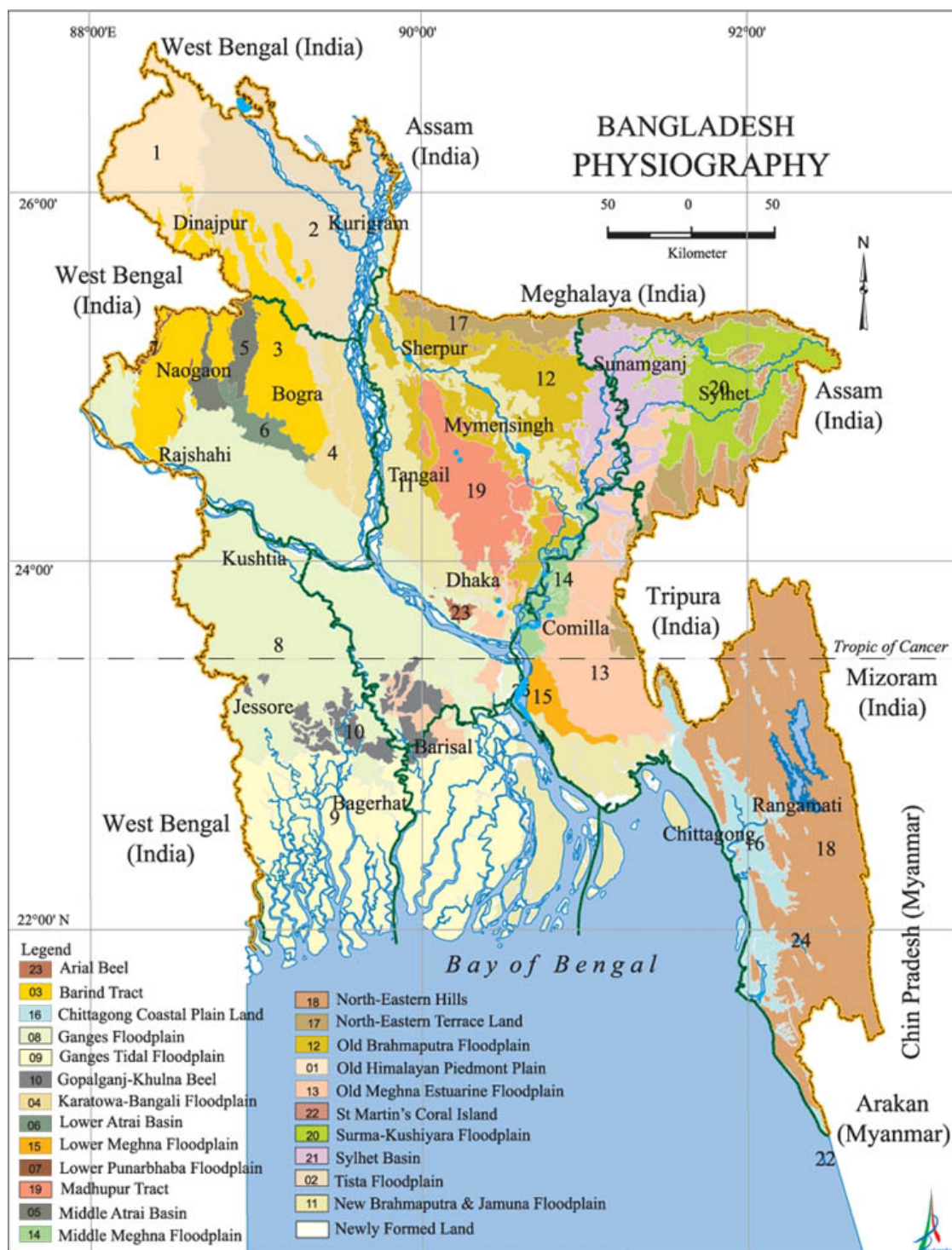


Figure 1.2: Physiographic map of Bangladesh.(Banglapedia)

1.5 Methodology

Samples were collected during the B.Sc (Hon's) part-III Geological fieldwork, 2021. The fieldwork was carried out through the traverse method and samples were collected from the field. These samples were selected for the analysis of major element composition (SiO_2 , Al_2O_3 , Fe_2O_3 , K_2O , MgO , P_2O_5 , Na_2O , CaO , TiO_2 , and MnOwt%). Collected samples were dried in the oven and crushed into powder form by an agate mortar. Aluminum caps (30 mm in dia) have been used for making pellets in these samples. Then the crushed samples were analyzed by X-ray fluorescence spectrometry (EDXL 300, Rigaku, Japan). Different models have been used to interpret provenance, tectonic setting, source area weathering and hydraulic sorting processes of the deposits using major oxides. $\text{SiO}/\text{Al}_2\text{O}_3\text{--Na}_2\text{O}/\text{K}_2\text{O}$ Diagram by Ohta, 2004. A–CN–K Diagram by Nesbitt and Young, 1984, Chemical Index of Alteration (CIA), Chemical Index of Weathering (CIW), Index of Compositional Variability (ICV) and Discriminant function multi-dimensional Diagrams given by Verma and Armstrong-Altrin, 2013; Roser and Korsch, 1988 have been used to understand the provenance, weathering and tectonic setting of the study area.